Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/14/2016
PI Name:	Hoffman, Larry Ph.D.		
Project Title:	Gravity-Induced Plasticity in Mammalian Utricular Hair Cells: Intrinsic or Multisensory?		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	90095-1624	**Congressional District:**	33
Comments:			
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No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 8/31/2018 per F. Hernandez/ARC (Ed., 7/13/17) NOTE: Extended to 7/31/2017, per NSSC information (Ed., 6/15/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schweizer, Felix (University of California, Los Angeles) Fuller, Charles (University of California, Davis)		
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Task Description:	This proposal is being submitted to the research topic area Ground based research designed to lead to space flight announced in NASA NRA (NASA Research Announcement) NNH12ZTT001N. We are proposing a series of studies to investigate whether the plasticity in the inner ear utricle is affected by non-vestibular load-sensing physiologic systems. Potential interactions between load-sensing systems might offer cross-modal synergy under altered gravity conditions, e.g., loading of the utricle might directly affect other systems (e.g., immune, osteogenic) that are currently subject to significant countermeasure efforts. In addition, the proposed research endeavors to elucidate how this plasticity is manifested in alterations of the activity of neurons that signal such changes to the central nervous system. We plan to use mouse models that systematically exclude the interaction of particular components of the vestibular load-sensing system. These models represent valuable adjuncts that can be implemented in future flight studies to address countermeasures driven by the recruitment of multiple sensory systems. Furthermore, the present investigation serves as a segue between altered gravity exposure durations that have been focused upon the duration of space shuttle missions and the much longer exposures incumbent with experiments aboard the International Space Station.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research explores the natural capabilities of utricular hair cells to undergo modifications in their synaptic connections with neurons that communicate sensory signals to the central nervous system as a result of exposure to altered gravitational fields. The ambient gravitational field is a fundamental component of the natural sensory environment, hence a principal aspect of our research is to explore the "sensory learning" that may result when changes in gravity are imposed. These fundamental findings are the foundation for future experiments to reveal the cellular mechanisms responsible for these changes. Since synaptic decline may reflect one form of natural vestibular dysfunction due to aging, pathology, or ototoxicity, an understanding of this repertoire and the cellular mechanisms is the first step toward designing rehabilitative strategies. Our research is designed to develop a systematic laboratory protocol in which these natural capabilities can be explored.
Task Progress:	The principal research achievements made during this funding year (August 1 2015 – July 31 2016) include: 1) improving breeding efficacy of otoferlin knockout mice (Otof-/-); 2) progress in implementing a slice preparation for measurement of hair cell capacitance and exocytotic ca-pacity; and 3) measurement of synapse distributions in adult and neonatal mice. Breeding colony for Otof-/- mice. We continued to experience difficulties in the breeding of Otof-/- mice through this grant period, without adequate explanation as to the reasons for the difficulties. Therefore, we deemed it necessary to adopt an alternative approach to the breeding. Subsequent to consultation with the mouse breeding laboratory in our vivarium we decided to generate females heterozygous for the Otof-/- mutation. We then requested (and thankfully received) two Otof-/- males from our colleagues at Johns Hopkins University with which we could mate with heterozygous females. In addition, we have implemented changes in the diet fed to pregnant dams in hopes this will lead to increased fecundity. We have delayed procuring the Nox3het animals until breeding of the Otof-/- animals was fully developed, as we anticipate that breeding of the Nox3het animals should proceed quickly. Since we need to optimize centrifugation runs, we deemed it best to do conduct the centrifugations at the same time, rather than in serial fashion. Exocytotic capacity in murine utricular hair cells. We are working on modifying the in vitro preparations for recording exocytotic capacity in mouse utricular hair cells. One of the difficulties in these measurements is recording from adult specimens, and establishing the direct link between synaptic modifications in specimens ex-posed to altered gravitational fields. Measurements of hair cell capacitance, as a measure of exocytotic capacity, has only been achieved in neonatal specimens due to the fact that adult specimens do not survive in whole mount in vitro conditions. To this end, we have implemented the utricular slice preparation. We are optimistic that this will be favorable for making capacitance measures from type I and II hair cells from all regions of the utricular epithelium, particularly in adult specimens. Synapse distribution in neonatal and adult mice. The distribution of synapses in utricular hair cells from wild-type (WT) and otoferlin-null (Otof-/-) mice was investigated using methods to immunolabel pre- and post-synaptic targets. We used anti-CtBP2 (C-terminal binding protein 2) to label the presynaptic ribbon and anti-GluR2 (glutamate receptor subunit 2) to identify the postsynaptic receptor. These studies parallelled the general strategy utilized to investigate synaptic plasticity in utricular hair cells in mice flown aboard STS-131 (Sultemeier et al., submitted), and provide the foundation for evaluating synaptic changes concomitant with hypergravity exposure. We defined a quantifiable synapse as a locus of close apposition of both CtBP2 and GluR2 immunopositive puncta. Type I hair cells were distinguished by the presence of postsynaptic calyces immunolabeled with anti-ß3-tubulin. Actin-rich stereocilia were labeled with fluorophore-conjugated phalloidin, which was used to navigate utricular topography (e.g., the line of polarity reversal). We then utilized semi-automated techniques to identify and quantify synaptic sites in specified topographic regions of the striola and extrastriolar regions of neonatal and adult mouse utricles. Synapse density was calculated as the ratio of synapse and hair cell counts. The ratio of type I to II hair cells from the WT striola was found to be 1.82:1 while that from the otoferlin-null striola was 1.25:1. In the medial extrastriola, the ratio was 1.82:1 for the WT and 1.37:1 for the otoferlin-null. We found that otoferlin-null striola overall exhibits lower synaptic density than the WT. Specifically, there was a decreased number of synapses/hair cell ($p=0.000275$), fewer iso-lated CtBP2-immunolabeled ("CtBP2-only"- those that are not associated with GluR2 puncta) ($p=0.000113$), as well as a smaller amount of total CtBP2-immunolabeled puncta (both those associated and not associated with GluR2) ($p<0.001$) in type I hair cells of the otoferlin-null compared to those of the WT striola. There were no significant differences in type II hair cells between otoferlin-null and WT striola for all three aforementioned parameters. Regarding the medial extrastriola, we found that otoferlin-null medial extrastriola overall possesses lower syn-aptic density than the WT. Quantitative analysis reveal a reduced number of synapses/hair cell in both type I ($p<0.001$) and type II ($p<0.05$) hair cells of the otoferlin-null compared to those of the WT medial extrastriola. Moreover, fewer CtBP2-only ($p<0.001$) and total CtBP2-immunolabeled puncta ($p<0.001$) were counted in type I hair cells of the otoferlin-null compared to the WT medial extrastriola. No significant differences were detected in type II hair cells between the two genotypes for the two aforementioned parameters. These data implicate that the distribution of maturing ribbon-associated synapses may be to some extent dependent upon stimulus-evoked exocytosis (either directly or indirectly via afferent modulation and centrifugal efferent activity), as the otoferlin-null mice display fewer synapses and ribbons.

	Furthermore, we investigated synapse density in hair cells from neonatal and adult utricular macula in order to examine the developmental maturation of ribbon-associated synapses. Although a decreased number of total CtBP2-immunolabeled puncta was found in the neonatal compared to the adult striola (p=0.000837), there were no significant differences in the synapse density between neonatal and adult striola. Measurements were performed in the medial extrastriola as well but no statistical differences were observed between the two age groups for the aforementioned parameters. These findings suggest that experience-driven plasticity possibly requires or promotes the direct emergence of a greater number of ribbons as the animal matures. Reference Sultemeier DR, KR Choy, FE Schweizer, and LF Hoffman. Spaceflight-induced synaptic modifications within hair cells of the mamnmalian utricle. Journal of Neurophysiology (submitted).
Bibliography Type:	Description: (Last Updated: 10/05/2022)
Abstracts for Journals and Proceedings	Oikonomou KD, Choy KR, Lopez IA, Schweizer FE, Hoffman LF. "Analysis of synaptic density of utricular hair cells in neonatal wild-type and Otoferlin deficient knockout mice." Presented at the 39th Annual MidWinter Meeting of the Association for Research in Otolaryngology, San Diego, CA, February 20-24, 2016. Abstracts of the Association for Research in Otolaryngology. 39th Annual MidWinter Meeting of the Association for Research in Otolaryngology, San Diego, CA, February 20-24, 2016. , Feb-2016